Working Title for Paper 2

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Abstract

This paper studies how protected areas and land-titling interventions affect deforestation outcomes across grid cells in Colombia. The abstract should eventually state the question, the treatment definitions, the identification strategy, the main result, and the contribution in five to eight sentences.

Keywords: protected areas; land titling; deforestation; Colombia; difference-in-differences

Contents

1 Introduction

This section should do four things quickly.

1. State the substantive question.
2. Explain why protected areas and land titling matter for forest loss.
3. Preview the identification challenge.
4. State the paper's main findings and contributions.

Useful paragraph structure:

- Motivation and policy relevance.
- Why comparing protected areas and titling is useful.
- What data and empirical design allow you to do.
- Main findings.
- Contribution to the literature.
- Roadmap.

Items you will likely want to reference here:

- The total extent of protected areas and titled lands over time.
- The number of newly treated grid cells by year.
- One anchor result from the baseline specification.

2 Background and Literature

2.1 Institutional Background

Use this subsection to explain:

- how protected areas are designated in Colombia,
- how Afro, Indigenous, and Campesino titling processes differ,
- what legal rights each treatment conveys,
- why treatment timing is staggered and potentially endogenous.

2.2 Conceptual Channels

Possible channels to discuss:

- restrictions on land use,
- stronger local monitoring or collective governance,
- changes in tenure security,
- displacement or leakage,
- state presence and enforcement,
- anticipatory behavior before formal recognition.

2.3 Related Literature

Organize the literature review into a few buckets:

- protected areas and deforestation,
- land tenure and environmental outcomes,
- Indigenous or collective property rights,
- staggered-adoption causal inference,
- remote sensing and gridded forest-loss measurement.

This section should end by clarifying exactly what gap your paper fills.

3 Data and Measurement

3.1 Unit of Analysis

The unit of analysis is a grid cell by year panel. Explain:

- grid resolution,
- time span,
- spatial coverage,
- the forest-loss outcome construction.

3.2 Treatment Construction

Define the treatment layers clearly:

- protected areas from WDPA yearly snapshots,
- Afro titles,
- Indigenous titles,
- Campesino reserves,
- any combined land-title treatment if used.

Important details to document:

- data source and raw file provenance,
- geometry cleaning choices,
- overlap rule used to mark a cell as treated,
- how first-treatment years are assigned,
- how pre-panel treatment is handled,
- whether treatment status is absorbing by construction.

3.3 Descriptive Patterns

This is a natural place to insert the graphs generated in the pipeline.

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Re-run the analysis pipeline or copy the final figure into `paper/figures/`.

Figure 1: Protected area extent over time.

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`02_land_title_area_by_year.png`

Re-run the analysis pipeline or copy the final figure into `paper/figures/`.

Figure 2: Cumulative land-titled area over time, overall and by title type.

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Figure 3: Treated grid cells by treatment type.

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`02_titling_grid_cells_over_time.png`

Re-run the analysis pipeline or copy the final figure into `paper/figures/`.

Figure 4: New and cumulative treated grid cells over time for land-titling interventions.

You will probably also want a summary table with:

- number of treated cells by layer,
- first and last treatment year by layer,
- mean overlap share,
- number treated before the panel starts.

4 Empirical Strategy

4.1 Treatment Timing Notation

Let G_i^m denote the first year in which grid cell i becomes treated under treatment type $m \in \{\text{protected area}, \text{Afro}, \text{Indigenous}, \text{Campesino}\}$. Define the absorbing treatment indicator as

$$D_i^m = \mathbf{1}\{G_i^m\}, \tag{1}$$

with $D_i^m = 0$ for cells never treated under layer m.

If you need to describe the spatial assignment explicitly, a clean notation is

$$\omega_{ig} = \frac{\text{Area}(C_i \cap P_g)}{\text{Area}(C_i)}, \tag{2}$$

where C_i is grid cell i and P_g is polygon g. A cell is treated when ω_{ig} satisfies the overlap rule used in the data construction.

4.2 Baseline Two-Way Fixed Effects Specification

A compact baseline specification is

$$Y_i = \alpha_i + \lambda + \beta D_i^m + X_i' \gamma + \varepsilon_i, \tag{3}$$

where α_i are grid-cell fixed effects and λ are year fixed effects.

If you include no time-varying controls, say so explicitly and justify it.

4.3 Event-Study Specification

Because treatment is staggered, you will likely want an event-study graph and a specification of the form

$$Y_i = \alpha_i + \lambda + \sum_{k \neq -1} \beta_k \, \mathbf{1}\{G_i^m = k\} + X_i'\gamma + \varepsilon_i. \tag{4}$$

This section should explain:

- the omitted event-time category,
- the estimator you use for staggered adoption,
- how never-treated and already-treated units enter the comparison,
- whether you trim extreme leads and lags.

4.4 Identification Concerns

Write down the core threats:

- non-random placement of protected areas or titles,
- pre-trends,
- endogenous timing,
- spillovers or leakage,
- differential measurement error across treatment types.

4.5 Standard Errors and Inference

Decide and document:

- clustering level,
- any spatial correction,
- weighting choice,
- treatment of serial correlation.

5 Results

5.1 Baseline Estimates

This subsection should present the main treatment-effect estimates first. Keep the table design simple and interpretable.

5.2 Dynamic Effects

Use event-study figures here. Explain both the pre-trend evidence and the post-treatment dynamics.

5.3 Comparing Protected Areas and Titling

This is likely one of the paper's comparative contributions. Consider structuring the discussion around:

- extensive-margin differences in treated area,
- differences in treated-cell composition,
- differences in average treatment effects,
- differences in timing and persistence.

5.4 Heterogeneity

Potential heterogeneity margins:

- forest frontier intensity,
- baseline deforestation risk,
- protected area versus title subtype,
- remoteness,
- overlap with state presence or conflict.

6 Mechanisms and Robustness

6.1 Mechanisms

Use this section only if you have credible mechanism evidence. Otherwise keep it short and relabel as "Additional Evidence".

Possible mechanism outcomes:

- changes in the level of annual loss,
- changes in persistence of loss,
- differential effects by initial pressure,
- interactions with institutional or geographic constraints.

6.2 Robustness Checks

Likely checks to consider:

- alternative overlap thresholds,

- dropping cells treated before the panel start,
- alternative treatment aggregations,
- excluding overlapping treatment types,
- restricting the sample to comparable support,
- placebo treatment dates,
- alternative estimators for staggered treatment timing.

6.3 Threats to Interpretation

Be explicit about the limits of the design. This usually makes the paper stronger.

7 Conclusion

The conclusion should not just repeat the results. It should:

- restate the question and answer,
- summarize the main empirical takeaway,
- explain the policy relevance,
- acknowledge the main limitation,
- point to the next research step.

A Appendix

A.1 Appendix Roadmap

Good candidates for the appendix:

- additional institutional detail,
- extra maps of treatment coverage,
- variable-construction details,
- robustness tables,
- estimator details,
- data validation checks,
- appendix figures for each treatment layer.

A.2 Additional Notation

If the paper becomes notation-heavy, put a compact notation table in the appendix.

A.3 Data Construction Details

Document here:

- WDPA snapshot selection,
- year parsing rules for land-titling shapefiles,
- geometry-cleaning assumptions,
- matching from polygons to grid cells,
- how cumulative area series were computed for the figures.